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| **Fiscal Year:** | FY 2012 | **Task Last Updated:** | FY 04/26/2012 |
| **PI Name:** | Steen, Paul H Ph.D. | | |
| **Project Title:** | Manipulation of two-phase flow in low-gravity environments: drop-phase relocation by active wettability control | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Fluid physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Organization Name:** | Cornell University | | |
| **PI Address 1:** | Chemical and Biomolecular Engineering | | |
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| **City:** | Ithaca | **State:** | NY |
| **Zip Code:** | 14853-5201 | **Congressional District:** | 23 |
| **Comments:** | PI is deceased as of September 2020 | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2008 Fluid Physics NNH08ZTT002N |
| **Start Date:** | 06/01/2009 | **End Date:** | 05/31/2013 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: New end date is 5/31/2013 per NSSC (Ed., 4/18/2012)<br>NOTE: extension to 5/31/2013 is in process, per PI (Ed., 3/6/12) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Daniel, Susan ( Cornell University )<br>Hui, Chung Yuen ( Cornell University ) | | |
| **Grant/Contract No.:** | NNX09AI83G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The broad technological goal of this project is to control two-phase flow in low gravity with active wettability. The scientific goal is to move droplets in a directed fashion using passive and active surface-energy gradients. The latter would yield active-wettability control. Fundamental to both goals is the stability of the sessile drop, both natural and forced response. Accordingly, sub-tasks are identified: I: Sessile-drop stability; and II: Active wettability-control.<br>To electronically control wettability of a substrate surface in real-time would enable manipulation of two-phase flow in the absence of earth's gravity. Condensers and separators are a crucial part of life support and power systems. A proof-of-concept device for collecting drops on a surface in low-gravity is the goal. Motivation comes from earth-based observations that passive chemical gradients can propel condensing steam droplets to a collector with centimeter-per-second speeds. Note that thermal effects are not central to the mechanism. The direction of movement of a drop is biased by the chemical gradient, and the energy to propel the droplet is provided by plane-normal oscillations. The oscillations also provide fluctuations sufficient to de-pin the contact-line. It is proposed to develop a tunable gradient to serve as a controllable bias and to develop a local vibration source to create the fluctuations. The goal is the capability to move drops on-demand in any number of directions.<br><br>A scientific basis for the technology requires understanding the nature of the contact-line, contact-angle and spreading behavior. Predictions of which drop oscillations will be most effective in de-pinning the contact-line and which vibrations will be most effective in propelling the drops are being carried out. Experiments to test the predictions are being conducted. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The location of the liquid- and gas-phases in two-phase liquid/gas flows on Earth depends crucially on the strength of Earth's gravity. Many devices that perform well on Earth are unreliable for lunar and Mars missions. They falter in low-gravity. For example, the Total Organic Carbon [1] analyzer that works well on Earth is ‘currently not in use due to difficulty in bubble removal.’ Hence, monitoring the quality of on-board water supplies becomes a high risk (red-light) issue for lunar and Mars missions (risk 38 of the Bioastronautic Critical Path Roadmap (BCPR) [1]). This is not an isolated instance. Condensers are crucial to Rankine-cycle power plants [2], among other applications, and for regenerative Life Support systems (CO2 condenser/separators) [3].<br>Additionally, success of this research effort will likely lead to better earth-based two-phase flow systems, including condensers and other heat-exchange devices.<br><br>1. Center, J. S., Bioastronautic Critical Path Roadmap, Supporting Documents.<br><br>2. Lahey, R. T.; Dhir, V., Research in Support of the Use of Rankine Cycle Energy Conversion Systems for Space Power and Propulsion. NASA/CR-2004-213142 2004.<br><br>3. [NAP2000], Microgravity Research in Support of Technologies for the Human Exploration and Development of Space and Planetary Bodies. National Academy Press: 2000. |
| **Task Progress:** | I. Sessile-drop stability. Surface tension holds a droplet placed on a planar substrate (a sessile drop) in a spherical-cap shape. Gravity is neglected. The contact-angle is specified by the liquid/gas/solid triplet of materials. This is the base-state whose stability we probe. Disturbances to the shape generate inviscid motions. Contact-line motions are governed by the Hocking condition, which relates deviations from equilibrium contact-angle to contact-line speed using a mobility parameter. Our study of the stability of the sessile drop is an analog of the Rayleigh capillary drop problem, which has proved so useful to microgravity science. We have discovered a two-parameter family of exact solutions to the sessile drop problem. The parameters are drop volume (or contact-angle) and mobility parameter. Solutions can be characterized by the vibration spectrum. In contrast to the Rayleigh spectrum, we find that the sessile drop spectra split and mix. Splitting occurs due to azimuthal dependence – the sectoral and tesseral modes – and is analogous to splitting in atomic spectra. Mixing occurs when higher wavenumbers have lower energies which also occur in atomic spectra. Continuing the analogy to atomic spectra, the results can be presented as a periodic table. In summary, we have discovered a two-parameter family of periodic tables that characterize the sessile drop behavior. Perhaps most important for application are the regions in the parameter plane where rapid mixing occurs. Knowledge of these regions enables the design of active wettability-control. Key predictions are being tested in experiment.<br>II. Active wettability-control. Wettability control will use a large array of reconfigurable membrane-covered-droplets. Acting in unison, these balloons can toggle the surface from hydrophilic to hydrophobic and back again. Acting individually, these membranes can create patches of wetting (when retracted) and de-wetting (when exposed) regions. In this way, surface properties can be modified locally or regionally, on demand. To reach this vision of wettability control, the mechanical behaviors of thin, circular membranes (PDMS or latex rubber) are being tested. These membranes are inflated to form balloon-like shapes. PDMS (polydimethylsiloxane) balloons are limited to about 200 per cent strains while latex rubber balloons can reach well-above 400 per cent strains. The effort is to characterize the mechanical properties and to realize active-wettability with an array of controlled balloon surfaces. The long-term goal is an array of controlled surfaces (individually or in unison) with diameters of 100 microns, by which locally adjustable wettability can be achieved. |
| **Bibliography Type:** | Description: (Last Updated: 09/04/2025) |
| **Abstracts for Journals and Proceedings** | Chang CT, Bostwick JB, Daniel S, Steen P. "Mechanical and dynamical response of pinned drops and polymer balloons." Presented at the 64th Annual Meeting of the APS Division of Fluid Dynamics, Baltimore, MD, November 20-22, 2011.<br>Bulletin of the American Physical Society 2011;56(18):BAPS.2011.DFD.G4.10. , Nov-2011 |
| **Articles in Peer-reviewed Journals** | van Lengerich HB, Steen PH. "Energy dissipation and the contact-line region of a spreading bridge." Journal of Fluid Mechanics. 2012 Jul;703:111-41. http://dx.doi.org/10.1017/jfm.2012.19 , Jul-2012 |
| **Articles in Peer-reviewed Journals** | Slater DM, Steen PH. "Bifurcation and stability of n coupled droplet oscillators with Sn symmetry." SIAM Journal on Applied Mathematics 2011;79(4):1204-19. http://dx.doi.org/10.1137/100808940 , Apr-2011 |

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| **Dissertations and Theses** | Bostwick JB. "Stability of constrained capillary interfaces." Dissertation, Cornell University, May 2011. , May-2011 |
| **Dissertations and Theses** | Slater DM. "Nonlinear dynamics of coupled capillary-surface oscillators." Dissertation, Cornell University, August 2011. , Aug-2011 |